

Daanesh Potnuri

📞 717-686-6770 ✉ daaneshpotnuri.3690@gmail.com [🌐 LinkedIn](#) [Portfolio](#) [🐙 Github](#)

Summary

Full-stack engineer with 3+ years of experience building modern frontends, secure backends, and API-driven microservices. Proven track record of integrating AI/LLM architectures to scale systems and automate workflows, recently accelerating document processing speeds by 50%.

Education

The Pennsylvania State University

08/2023 - 05/2025

Masters of Science in Computer Science, CGPA: 3.36/4.0

Pennsylvania, USA

Experience

Software Developer – Ras Lavrar LLC

04/2026 – Present

- Overhauled a production legal-document automation workflow by implementing PA-specific OCR routing, affidavit intake, and ClientLookup tools, boosting file classification accuracy to 99% and reducing manual routing time by 40%.
- Transformed a legacy redaction pipeline by coding event-driven Django webhooks to autonomously parse client zip uploads and validate PDFs, accelerating document processing speed by 50% and eliminating 100% of manual extractions.
- Devised an AI-powered judgment processing pipeline utilizing Hugging Face NLP models to extract and validate data from complex legal PDFs, automating JST ingestion, saving 15+ hours weekly, and increasing extraction accuracy by 35%.
- Constructed a Python/PySide6 and MySQL desktop application to reconcile database records against network workflows, automating discrepancy detection and report generation, which reduced daily reconciliation efforts by 60%.

Software Developer – World Disaster Center

07/2025 – 04/2026

- Engineered a React and Node.js/Express CMS and searchable roster with server-side pagination, PostgreSQL integration, and JWT-secured APIs, increasing publishing throughput by 40% and cutting admin update time by 30%.
- Optimized backend performance by migrating to normalized PostgreSQL REST services with advanced indexing and integrating GA4 event analytics, improving median API response time by 25% and boosting user session duration by 20%.

Graduate Research Assistant & Software Developer – The Pennsylvania State University

08/2023 – 05/2025

- Architected a Python full-stack data labeling app (FastAPI, PostgreSQL/SQLAlchemy, JWT, Docker), automated CSV ingest and uploads, and orchestrated a custom admin dashboard—streamlined data prep time by 50%.
- Spearheaded a custom React + Node/Express portal with dynamic server-side pagination, advanced filters, and secure role-based views; optimized administrative workflows to reduce TA overhead by 30% and improve student turnaround.
- Designed and operationalized a Transformer + BiLSTM forecasting pipeline (NYC) in TensorFlow/Keras; benchmarked results (RMSE 34.1, MAE 23.7, MAPE 4.1%, R^2 0.92) and automated ETL, evaluation, and CI.

Associate Software Engineer – Mphasis

08/2020 – 08/2023

- Upgraded Java/Spring Boot REST services and Hibernate configurations for the Fircosoft sanctions platform, boosting data ingestion from 8M to 12M records/day, scaling throughput to 700 req/s, and reducing API latency by 26%.
- Orchestrated containerization for 12 core backend microservices using Docker/Kubernetes and automated seamless Jenkins CI/CD delivery pipelines, accelerating releases and slashing deployment failures by 40% and environment drift by 50%.
- Fortified overall system reliability by automating 200+ JUnit/Selenium QA tests and deploying real-time Splunk/Prometheus observability monitoring, maintaining 99.9% uptime and reducing incident response time by 35%.

Skills Summary

Languages: Python, Java, TypeScript/JavaScript, SQL

Frontend: React, Next.js, HTML/CSS, Tailwind; SPA state server-side pagination; form validation

Backend & APIs: Django, PySide6, OCR, Hugging Face, Spring Boot (REST, JPA/Hibernate), Node.js/Express, FastAPI; API design, JWT/RBAC, caching

Databases: PostgreSQL, Oracle, MySQL, MongoDB, Redis; schema design, indexing, query optimization; ORMs (Hibernate, SQLAlchemy)

Cloud DevOps: Docker, Kubernetes, Jenkins, GitHub Actions, AWS (EC2, S3, RDS, IAM), Vercel; CI/CD, env promotion, IaC (Terraform – basics)

Quality, Observability Security: JUnit, Selenium, PyTest, Postman; performance/load testing, coverage, regression; Splunk, Prometheus/Grafana, Sentry, GA4; OAuth2/JWT, RBAC, secrets mgmt, rate limiting, secure APIs; Git, Agile/Scrum, PR reviews, Swagger/OpenAPI, Jira

Projects

RAG-Based QA Bot | *PyTorch, DistilBERT, FAISS, FastAPI, LangChain, Ollama, Ngrok (DOM parsing)*

- Engineered a domain-specific RAG chatbot over the Amazon Reviews 2023 dataset (213K+ products) using DistilBERT, FAISS, and LangChain for retrieval and response generation. Integrated sentiment analysis and optimized retrieval to deliver accurate, real-time product insights with 2–3 s latency and 95% off-topic query filtering.

Decentralized File Sharing dApp | *Blockchain, Solidity, React.js, IPFS*

- Developed a secure peer-to-peer file-sharing dApp on Ethereum, implemented Solidity contracts for access control and metadata, and integrated IPFS (Pinata) with a React + Web3.js frontend—accelerating onboarding by 80%, maintaining 99.99% uptime, and validating smart-contract security with 95% test coverage.

Automated Examination System | *Python, Flask, NLP, Machine Learning*

- Operationalized an NLP pipeline to auto-generate and grade multiple-choice questions, integrated into a Flask web app for instructor review/edit/export—IJITEE (2020) publication—cutting manual authoring time by 30%.